

Pandora’s Box: Cross-Chain Arbitrages in the Realm of Blockchain Interoperability

Burak Öz
Technical University of Munich
Flashbots

Christof Ferreira Torres
Instituto Superior Técnico
University of Lisbon
INESC-ID

Jonas Gebele
Technical University of Munich

Filip Rezabek
Technical University of Munich

Bruno Mazorra
Flashbots

Florian Matthes
Technical University of Munich

Abstract

Over recent years, the blockchain ecosystem has grown significantly with the emergence of new Layer-1 (L1) and Layer-2 (L2) networks. These blockchains typically host Decentralized Exchanges (DEXes) for trading assets such as native currencies and stablecoins. While this diversity enriches the ecosystem, it also fragments liquidity, posing challenges for DEXes offering the same assets across multiple blockchains. This fragmentation leads to price discrepancies, creating opportunities like arbitrages for profit-seeking traders, which fall under the broader category of exploitative economic practices known as Maximal Extractable Value (MEV). Although MEV extraction has been extensively studied within single domains (i.e., individual blockchains), cross-chain arbitrages, a form of cross-domain MEV, have received little attention due to their non-atomic nature, complicating both execution and detection.

In this paper, we shed light on opaque cross-chain MEV activities by presenting the first systematic study of two non-atomic cross-chain arbitrage strategies: Sequence-Independent Arbitrage (SIA) and Sequence-Dependent Arbitrage (SDA). The former involves independent, opposite-direction trades across chains, while the latter relies on asset bridges. We analyze the effectiveness of these strategies across nine blockchains over a one-year period from September 2023 to August 2024, identifying 260,808 cross-chain arbitrages, 32.37% of which involve bridging solutions. These arbitrages generated a lower-bound profit of 9,496,115.28 USD from a total traded volume of 465,797,487.98 USD. Additionally, we examine the security implications of cross-chain arbitrages, uncovering centralization among arbitrageurs, network congestion caused by failed transactions, and growing private mempool adoption. Finally, we discuss sequencer incentives and propose a risk-optimized arbitrage strategy.

1 Introduction

The blockchain ecosystem has expanded rapidly in recent years, partially driven by a surge in Layer-2 (L2) solutions designed to scale blockchain execution through lower execution costs and higher transaction throughput. Decentralized Finance (DeFi) applications remain at the forefront of this growth, with cumulative daily trading volumes exceeding 7 B USD and Total Value Locked (TVL) surpassing 100 B USD [16]. This high activity and volume enables value extraction across various blockchains, creating lucrative opportunities such as arbitrages, liquidations, and sandwiches—collectively known as Maximal Extractable Value (MEV) [15]. On Ethereum alone, it is estimated that 14.23 B USD were already extracted via atomic arbitrages between September 2023 and August 2024 [27].

However, MEV is not confined to a single domain due to the interoperability between blockchains [33] and its intersections with non-blockchain domains like Centralized Exchanges (CEXes) [26]. Obadia et al. define a *domain* as a system with a shared state, such as a blockchain, modified by actions like transactions [33]. Within a domain and a given state transition window (e.g., a series of blocks), MEV represents the maximum value a user with control over transaction sequencing, insertion, and censorship can extract [33]. Extending this concept to ecosystems with multiple domains, they introduce *cross-domain MEV* as the maximum cumulative balance increase a user can achieve by controlling transaction sequencing across n blockchains [33].

Prior research highlights the security threats MEV poses to blockchains, including consensus vulnerabilities, network congestion, centralization risks, and censorship [15, 35–37, 40, 49, 50]. However, these studies primarily focus on single-domain MEV, which is easier to detect than cross-domain MEV due to the latter’s non-atomic nature. Cross-domain MEV involves matching activities across various blockchains with asynchronous timelines (e.g., differing block times) and actors with potentially distinct identities, making detection more complex. Consequently, empirical studies on cross-

* All links are valid as of 22nd January, 2025

domain MEV, particularly from a security perspective, remain sparse, apart from recent work on non-atomic arbitrages between CEXes and Decentralized Exchanges (DEXes) [26].

Our work presents the first systematic empirical study on cross-domain MEV extraction, focusing specifically on *cross-chain arbitrages*. Many blockchains host DEXes that offer identical or similar versions of assets (e.g., wrapped or bridged tokens), where price discrepancies arise due to asynchronous states. These discrepancies enable cross-chain arbitrage by executing transactions on individual blockchains to exploit price differences. However, unlike atomic, single-chain arbitrages, cross-chain arbitrages require efficient inventory management across multiple blockchains or the use of bridging technologies, both of which introduce risks to successful execution [33] and may hinder adoption.

Furthermore, CEX-DEX arbitrages conducted between individual blockchains and CEXes are arguably more profitable than cross-chain arbitrages due to the deeper liquidity and lower fees offered by CEXes, such as Binance [10], raising doubts about the viability of the latter. Nonetheless, advances in blockchain interoperability have potentially made cross-chain arbitrages more feasible, lowering the barrier to entry for those without inventory on multiple networks. Thus, we hypothesize that the significance of cross-chain arbitrages as a source of MEV has increased over time with the emergence of cross-chain bridges, although empirical evidence and associated security threats remain largely unexplored.

To shed light on the currently opaque cross-chain MEV activities performed by users, we analyze cross-chain arbitrages between September 2023 and August 2024 across nine different blockchains, including both Layer-1 (L1) and L2 networks. We examine the frequency and profitability of these arbitrages, identifying the most arbitrated and profitable blockchain pairs. We also evaluate arbitrage durations, given the time-sensitive nature of profits, and compare tokens used in cross-chain arbitrages to those in CEX-DEX arbitrages. Additionally, we examine bridge usage to understand arbitrage execution strategies. To assess the security implications, we investigate arbitrageur centralization, network congestion, and the use of private mempools. Finally, we discuss how cross-chain arbitrages can incentivize controlling sequencing across multiple blockchains and present a risk-optimized arbitrage strategy.

Contributions. We make the following contributions:

- We propose a blockchain-agnostic methodology for detecting cross-chain arbitrages and identifying cross-chain bridge transactions.
- We conduct the first systematic study measuring the prevalence of cross-chain arbitrages in the wild over one year, spanning nine different blockchains. Our analysis uncovers 260,808 cross-chain arbitrages, of which 32.37 % are executed using bridging solutions.

- We demonstrate that cross-chain arbitrages are profitable, with a lower-bound total profit of approximately 9,496,115.28 USD from a total traded volume of 465,797,487.98 USD. Additionally, we identify potential security threats, including centralization among cross-chain arbitrageurs, significant amount of failed transactions by arbitrage bots, and an upward trend in private mempool usage.

2 Background

This section provides relevant background on DeFi, MEV, L1 scaling solutions, and blockchain interoperability.

2.1 Decentralized Finance

DeFi offers similar primitives as Traditional Finance (TradFi) but extends it via additional solutions that are solely enabled by blockchain technology. These include DEXes, lending platforms, options markets, and tokenized assets, all implemented via smart contracts. While CEXes use a Central Limit Order Book (CLOB) model, which relies on a centralized entity for order matching and settlement, DEXes typically adopt Automated Market Makers (AMMs) to facilitate Peer-to-Peer (P2P) trading. AMMs rely on liquidity pools and predefined mathematical formulas, such as the constant product market maker (Constant Product Market Maker (CPMM)), to determine trade prices and execute transactions without intermediaries. For example, Uniswap [45] employs CPMM to maintain price stability and enable automated price discovery. Trading on DEXes introduces price slippage—both expected, due to trade volume and liquidity constraints, and unexpected, due to execution delays and market volatility—leading to measurable market impacts. This slippage and other arbitrage opportunities are relevant to arbitrageurs, which monitor various blockchains with the aim to extract monetary value.

2.2 Maximal Extractable Value

The decentralized infrastructure enables opportunities for MEV extraction. MEV relies on two key transaction ordering primitives: frontrunning, where an extractor ensures their transaction precedes a target transaction (T_{Target}), and backrunning, where the extractor’s transaction executes immediately after T_{Target} . Common MEV strategies include arbitrage, liquidation, and sandwiching. Arbitrage exploits price discrepancies across exchanges by analyzing blockchain state changes, ensuring price alignment across DEXes. Liquidations involve repaying a debt to purchase discounted collateral, often focusing on fixed-discount opportunities that can be executed in a single transaction. Both arbitrage and liquidation are generally considered beneficial for market efficiency. In contrast, sandwiching is a manipulative strategy where an adversarial transaction wraps a target trade (T_V), buying the

asset beforehand to profit from the price increase caused by T_V , and subsequently selling at a higher price. This practice disrupts fair-price execution and is considered harmful.

Competition among MEV extractors may cause block congestion and gas fee inflation. MEV can also result in systemic instability by incentivizing behaviors prioritizing individual gain over network security. In the context of blockchain, this can lead to consensus instabilities, as miners or validators may be incentivized to steal transactions from others by creating forks, thereby undermining the system’s integrity. Among the various strategies for MEV extraction, arbitrage is the most prevalent, as demonstrated in measurement studies in [15, 22, 37].

2.3 Layer-1 and Scaling Solutions

L1 refers to the classical blockchain model where transactions are recorded on a public, immutable, and trustless ledger which follows a consensus algorithm to determine block creation. Examples of popular L1s include Ethereum [47], Binance Smart Chain [11], Avalanche [39], etc. However, L1s face severe scalability issues in terms of throughput and transaction fees. As a result, a number of so-called L2 scaling solutions have emerged, where the most prominent solutions are either based on sidechains (e.g., Polygon) or commit chains (e.g., rollups). Commit chains or rollups can be further split into either optimistic (e.g., Arbitrum) or zero-knowledge based (e.g., Scroll) depending on what types of proofs are used to verify the validity of L2 transactions. Sidechains typically run a much faster consensus mechanism among fewer peers in parallel to L1. On the other hand, rollups enable throughput scaling by off-loading compute and (possibly) storage resources off-chain without a need for large-scale consensus, which is provided by the underlying L1. Generally, L2 scaling solutions allow distrustful parties to deposit funds into a bridge smart contract on L1 and then operate on L2 via L2 transactions whose state is then updated to L1.

Table 1 summarizes the various blockchains that we analyze for cross-chain arbitrages. Unlike L1s, which typically operate using a gas price model and provide public access to

Blockchain Name	Type	Mempool	Ordering	Block Time	Launch
Avalanche (AVA)	L1	Public	Gas Price	~2.00 s	Sep 2020
Binance Smart Chain (BSC)	L1	Public	Gas Price	~3.00 s	Sep 2020
Ethereum (ETH)	L1	Public	Gas Price	~12.00 s	Jul 2015
Polygon PoS (POL)	L2-SC	Public	Gas Price	~2.00 s	Jun 2020
Arbitrum (ARB)	L2-OR	Private	FCFS*	~0.25 s	Aug 2021
Base (BASE)	L2-OR	Private	FCFS*	~2.00 s	Aug 2023
Optimism (OP)	L2-OR	Private	Gas Price*	~2.00 s	Dec 2021
Scroll (SCROLL)	L2-ZK	Private	FCFS*	~3.50 s	Oct 2023
zkSync Era (ZKSYNC)	L2-ZK	Private	FCFS*	~1.00 s	Mar 2023

* Centralized sequencers.

Table 1: Overview of investigated Layer-1 (L1) and Layer-2 (L2) solutions. SC - Sidechain, OR - Optimistic Rollup, ZK - ZK-Rollup, FCFS - First-Come, First-Served.

mempool data, L2s typically feature private mempools with transactions ordered by centralized sequencers, which often follow a First-Come, First-Served (FCFS) strategy. The latter makes it harder for users to extract MEV as they cannot observe other pending transactions and cannot pay higher gas fees to prioritize their own transactions. The block time does not correspond to the finality, as L2s only offer a soft finality to its users as the transactions are only fully finalized after being settled on L1.

2.4 Bridges and Blockchain Interoperability

Blockchain interoperability aims to connect L1 and L2 blockchain networks and transfer assets or data between them. The transfer typically involves locking assets on one chain and minting an equivalent representation on the destination chain. For example, a token can move from one blockchain to another while maintaining security guarantees. Bridges are the basic infrastructure enabling this interoperability and can be categorized into either *native* or *multichain* bridges. Each bridge solution comes with different security, latency, and cost trade-offs.

Native bridges are built into the architecture of the underlying blockchain and facilitate direct transfers between their L1s and L2s. The process typically involves asset locking via a smart contract and proof generation of the locked asset, and transmission of it to the destination chain. On the destination chain, an equivalent amount of the asset is either minted (if the bridge uses wrapped tokens) or unlocked. Multichain bridges, on the other hand, are more versatile than native bridges because they can support several different blockchains, while native bridges only operate between two blockchains. Multichain bridges follow the same steps of locking, proof generation, and minting/unlocking on the other chain, except that they present a unified communication mechanism that is blockchain-agnostic.

3 Methodology

In this section, we outline our methodology for detecting and analyzing cross-chain arbitrages as well as identifying native and multichain bridge transactions.

3.1 Cross-Chain Arbitrage Detection Model

We model a system of blockchain networks $\mathcal{B}$, where each blockchain $b \in \mathcal{B}$ hosts a set of transactions T_b . The aggregated set of transactions across all blockchains is:

$$\mathcal{T} = \bigcup_{b \in \mathcal{B}} T_b.$$

We define $S_b \subset T_b$ as the set of transactions on blockchain b involving at least one *swap* action on a DEX. The aggregated

set of all swap transactions is:

$$\mathcal{S} = \bigcup_{b \in \mathcal{B}} \mathcal{S}_b, \quad \text{with } \mathcal{S} \subset \mathcal{T}.$$

A swap transaction $s \in \mathcal{S}$ consists of k sequential swap actions. Each action involves:

- An input asset α^i ;
- An output asset α^o ;
- Input amount $\pi(\alpha^i)$;
- Output amount $\pi(\alpha^o)$.

Each transaction s can be summarized by its first input asset $\alpha_{s,1}^i$ and final output asset $\alpha_{s,k}^o$, where k is the total number of swap actions.

To reduce complexity and search space, in this study, we focus on detecting *2-hop* cross-chain arbitrages, involving pairs of swap transactions s_m and s_n on different blockchains $m, n \in \mathcal{B}$ ($m \neq n$). We define the following key functions for these pairs:

- **Marginal Difference (δ):** Measures the relative difference in amounts between the input and output assets of the transactions, indicating hedging efficiency:

$$\delta(s_m, s_n) = \min \left(\frac{|\pi(\alpha_{s_m}^i) - \pi(\alpha_{s_n}^o)|}{\pi(\alpha_{s_n}^o)}, \frac{|\pi(\alpha_{s_n}^i) - \pi(\alpha_{s_m}^o)|}{\pi(\alpha_{s_m}^o)} \right).$$

- **Time Difference (λ):** The time interval between the two transactions.

A *2-hop* cross-chain arbitrage match is detected if the following conditions are satisfied:

$$\mathcal{A} = \{(s_m, s_n) \in \mathcal{S} \mid m \neq n, \alpha_{s_m}^i \equiv \alpha_{s_n}^o, \alpha_{s_m}^o \equiv \alpha_{s_n}^i, \delta(s_m, s_n) \leq \Delta, \lambda(s_m, s_n) \leq \Lambda\}.$$

Here:

- $\equiv$: Accounts for equivalent assets (e.g., $USDC \equiv USDT$ or $ETH \equiv WETH$) by applying necessary conversions.
- Δ : Maximum allowable marginal difference.*
- Λ : Maximum allowable time difference.

Stricter (Δ, Λ) thresholds apply to swap transactions involving stablecoin or native currency pairs, such as $WETH$ - $USDC$ (denoted as $\mathcal{P}$), since these pairs are highly volatile and are typically used for CEX-DEX arbitrage due to the deeper liquidity and lower fees offered by CEXes. For other pairs, we

* This condition is applied exclusively to the hedge asset.

allow a longer time difference to account for cross-chain arbitrages involving bridge usage, which may require extended completion times. Specifically:

$$(\Delta, \Lambda) = \begin{cases} (0.01, 3600s), & \text{if } (\alpha_s^i, \alpha_s^o) \notin \mathcal{P}, \\ (0.001, 30s), & \text{otherwise.} \end{cases}$$

In the $\notin \mathcal{P}$ case, we allow a relaxation of Δ to 0.05 if $\Lambda \leq 60s$, as such cross-chain arbitrages often bypass bridges and rely on existing inventory, which may lack the precision of bridging exact amounts.

For each $s \in \mathcal{S}$, the set of matches is defined as:

$$C_s = \{(s, s') \in \mathcal{A} \mid s' \in \mathcal{S}\}.$$

Since each transaction can be involved in at most one cross-chain arbitrage, the *best match* is identified by eliminating false positive matches using a selection function ϕ that minimizes the marginal difference $\delta(c)$:

$$\phi(C_s) = \arg \min_{c \in C_s} \delta(c).$$

The cumulative set of all detected *2-hop* cross-chain arbitrages across $\mathcal{B}$ blockchain networks is the union of the best matches for all $s \in \mathcal{S}$:

$$\mathcal{M}_{\mathcal{B}} = \bigcup_{s \in \mathcal{S}} \{c^* \mid c^* = \phi(C_s), C_s \neq \emptyset\}.$$

3.1.1 Profit Calculation

To assess the profitability of a detected cross-chain arbitrage, we define a profit function $P(s_m, s_n)$ for a pair of swap transactions s_m and s_n as:

$$P(s_m, s_n) = \pi(\alpha_{s_m}^o) - \pi(\alpha_{s_n}^i) - \rho(s_m, s_n),$$

where:

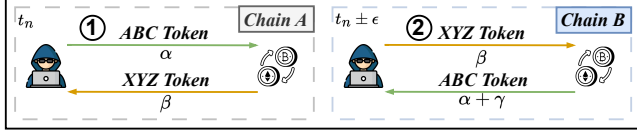
- $\pi(\alpha_{s_m}^o)$: Output amount from the profit leg (s_m).
- $\pi(\alpha_{s_n}^i)$: Input amount from the hedge leg (s_n).
- $\rho(s_m, s_n)$: Total fees, including gas fees, coinbase payments*, and bridging costs.

3.2 Arbitrage Strategies

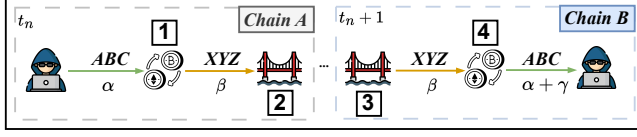
We define two generic cross-chain arbitrage strategies based on their order of execution:

Sequence-Independent Arbitrage (SIA). In SIA, the arbitrageur maintains an inventory on both chains. The execution of arbitrage legs does not necessarily follow a strict order (i.e., the profit leg can occur before the hedge

* A coinbase payment is a direct ETH transfer to the coinbase address of an Ethereum block builder.



(a) Sequence-Independent Cross-Chain Arbitrage (SIA).



(b) Sequence-Dependent Cross-Chain Arbitrage (SDA).

Figure 1: Cross-chain arbitrage strategies. Figure 1a illustrates an SIA where the arbitrageur executes arbitrage legs without a strict sequence. Specifically, the hedge leg is executed on Chain A at time t_n , while the profit leg is executed on Chain B within a close time window, $t_n \pm \epsilon$. In contrast, Figure 1b demonstrates an SDA, where the arbitrageur executes the hedge leg and sends assets to the bridge on Chain A at t_n . The assets are received and the profit leg is executed on Chain B at $t_n + 1$, showing that the profit leg depends on the hedge leg and is executed sequentially.

leg), allowing the arbitrageur to independently submit transactions. Since the output of the hedge leg is not directly required as input for the profit leg, unlike atomic arbitrages where swap outputs feed into subsequent swaps, the sequence of transactions is flexible. However, with high probability, the continuous execution of SIAs in the same direction drains inventory over time. To address this, arbitrageurs are required to rebalance their inventories either immediately after an arbitrage or at regular intervals.

Figure 1a illustrates a generic SIA flow, where an arbitrageur swaps α ABC tokens for β XYZ tokens on Chain A (①) and swaps β XYZ tokens for $(\alpha + \gamma)$ ABC tokens on Chain B (②). The profit is γ minus any fees that were paid to perform the swap transactions on both chains. Please note that the relative execution order of transactions does not matter as the arbitrageur holds enough funds on both chains to perform the swaps independently, meaning that the swap on Chain B could have been performed prior to the swap on Chain A.

Sequence-Dependent Arbitrage (SDA). In SDA, the arbitrageur transfers the output of the hedge leg to the other blockchain via a native or a multichain bridge. The bridged assets are then used as input for the profit leg. As the successful execution of this strategy requires sequential processing of transactions across blockchains, we classify it as sequence-dependent.

Figure 1b presents a generic SDA flow, where an ar-

Transaction Hash	Blockchain	From	To	Type	Time
0xeb546...37f7c	Base	0x4cb...3d9	0x6bd...891	Swap	12:25:51 AM
0xffefb...aa7aa	Arbitrum	0x4cb...3d9	0x1b0...506	Swap	12:26:18 AM

Table 2: Example of an SIA between Base and Arbitrum.

Transaction Hash	Blockchain	From	To	Type	Time
0x59237...61999	Ethereum	0xe6b...532	0x111...a65	Swap	11:13:11 AM
0x64fa3...46aa8	Ethereum	0xe6b...532	0xa7c...44d	Bridge	11:14:11 AM
0x73307...591ec	BSC*	0xe6b...532	0xa7c...44d	Bridge	11:19:10 AM
0x101d0...cea91	BSC*	0xe6b...532	0x111...a65	Swap	11:19:22 AM

* Binance Smart Chain.

Table 3: Example of an SDA between Ethereum and BSC.

bitrageur first swaps α ABC tokens for β XYZ tokens on Chain A (①), then performs a bridge transaction to transfer XYZ tokens from Chain A to Chain B (②-③), and finally swaps β XYZ tokens for $(\alpha + \gamma)$ ABC tokens on Chain B (④). The profit is γ minus any fees that were paid to perform the swap transactions on both chains as well as the bridge transactions. Please note that the relative execution order of transactions does matter as the arbitrageur does not maintain enough inventory on both chains to perform the swaps independently.

3.2.1 Case Study: SIA between Base and Arbitrum

Table 2 presents a real-world example of an SIA. The arbitrageur first swaps 0.9 WETH for 1,034,616.49 OMNI on Base at 12:25:51 AM. On Arbitrum, they swap 1,034,616.49 OMNI for ~ 1 WETH, earning approximately 198.95 USD in profit after deducting transaction fees. The entire SIA was completed in 27 seconds. Post-arbitrage, the arbitrageur executes a rebalancing strategy by transferring 1,034,616.49 OMNI from Base to Arbitrum using the LayerZero bridge.

3.2.2 Case Study: SDA between Ethereum and BSC

Table 3 presents a real-world example of an SDA. The arbitrageur initially swaps 20,000 USDT for 2,692,596.48 VOW on Ethereum at 11:13:11 AM. Following this, they approve DBridge to transfer VOW tokens and deposit 2,692,590 VOW 1 minute later. The arbitrageur receives 2,692,590 VOW on Binance Smart Chain at 11:19:10 AM and, 12 seconds later, swaps 2,692,593.08 VOW for 134,889.18 Binance-Peg BSC-USD, earning $\sim 114,688.38$ USD in profit after transaction fees. The entire SDA took over 6 minutes to complete.

3.3 Bridge Usage Detection

To gauge what strategy (i.e., SIA or SDA) arbitrageurs are employing, we need to leverage a mechanism to separate

* Rebalance transactions: Base 0x11e...dcf, Arbitrum 0x2b8...7dca.

the two strategies. From a detection point of view both SIA and SDA will be simply flagged as cross-chain arbitrages by our methodology introduced in Section 3.1. However, what clearly sets both strategies apart is the fact that arbitrageurs following the SDA strategy need to bridge their assets from one chain to another in order to finalize their arbitrage. Hence, detecting whether arbitrageurs performed bridge transactions in between the swaps allows us to mark a cross-chain arbitrage as either SDA or SIA.

In practice there are two types of popular bridges that arbitrageurs can leverage: native bridges and multichain bridges. Table 4 provides an overview of the detected bridges in this work as well as bridge type, average bridging time, and detection method. Some bridges serve as an infrastructure on which other bridges can be built, as is the case of LayerZero and Stargate. We employ two distinct methods to link and detect bridge transactions between cross-chain arbitrages:

Unique Identifier. This method leverages unique identifiers that are typically emitted by native bridges whenever cross-chain bridge transactions are performed. These identifiers allow us to link cross-chain bridge transactions even if assets are sent between different sender and receiver addresses. The detection is split into two phases where in the first phase we create a dataset of all native bridge transactions that were performed during the period of our study. In a second phase we check whether arbitrageurs used native bridge transactions in between their arbitrage legs.

For the first phase we borrow the method from [22] and adapt it to link native bridge transactions on Arbitrum, Base, Optimism, Scroll, and zkSync. We build the dataset by scanning Ethereum and the respective rollups for events triggered by their native bridge smart contracts deployed by the individual rollups (i.e., Arbitrum, Base, Optimism, Scroll, and zkSync). For example, for Arbitrum we scan for the “InboxMessageDelivered” event on Ethereum and the corresponding “RedeemScheduled” event on Arbitrum. Transactions on both sides are then linked together using the “message number”, a unique sequential identifier generated via

Arbitrum’s bridge smart contract. Please note that we only consider native bridge transactions that go from Ethereum to rollups and not vice versa. This is because native bridging from Ethereum to a rollup takes typically a few minutes whereas bridging from a rollup to Ethereum via a native bridge takes typically several days, making the later impractical for cross-chain arbitrages as the price differences will be most likely not available anymore.

After building a dataset of native cross-chain transactions, we check in a second phase whether we can find a pair of native cross-chain bridge transactions that are performed by the same address of the cross-chain arbitrage and within the time interval of the cross-chain arbitrage, thereby leveraging the same token that was the output of the swap of the first leg of the arbitrage and the input of the swap of the second leg of the arbitrage. If the aforementioned criteria is met, we label the cross-chain arbitrage as SDA, otherwise as SIA. Please note that for Polygon, we cannot employ the unique identifier method as its native bridge does not leak any unique identifiers. For Polygon and multichain bridges, we leverage the token transfer method.

Token Transfer. In this method we link bridge transactions via the ERC-20 token transfer events that are emitted as part of locking and minting tokens on either side of the bridge. This provides a less efficient mechanism to link bridge transactions but also a more generic one that works with bridges that do not leak a unique identifier. The idea is to search for token transfer events emitted in between the first leg of the arbitrage and the second leg of the arbitrage and to link them together through identical sender and receiver addresses from the detect cross-chain arbitrage.

We start by first searching on a block per block basis on the source chain (i.e., where the first arbitrage leg was performed) for transfer events emitted by the same token as the last token swapped in the first leg of the arbitrage. We stop our search the moment we find a token transfer event where the source of the transferred tokens is the same address as which received the swapped tokens in the first leg of the arbitrage. Next, we perform a reverse search on the destination chain (i.e., where the second arbitrage leg was performed) by starting at the block of the second arbitrage leg and searching previous blocks, on a block per block basis, for transfer events emitted by the same token as the first token swapped in the second leg of the arbitrage. We stop our search the moment we find a token transfer event where the destination of the transferred tokens is the same address which initiated the token swap in the second leg of the arbitrage.

If we find two token transfer events, one on each chain, meeting the aforementioned criteria, we assume that both transfer events belong together and mark their originating transactions as cross-chain bridge transactions and hence the cross-chain arbitrage as SDA and otherwise as SIA.

Bridge Name	Bridge Type	Bridging Time	Detection Method
Across [3]	Multichain	1-4 mins	Token Transfer
Arbitrum One [6]	Native	15-30 mins	Unique Identifier
Axelar [9]	Multichain	up to 1 hour	Token Transfer
Base [43]	Native	~10 mins	Unique Identifier
Celer [12]	Multichain	5-20 mins	Token Transfer
LayerZero [42]	Multichain	~4 mins	Token Transfer
Optimism [34]	Native	1-3 mins	Unique Identifier
Polygon [28]	Native	10-30 mins	Token Transfer
Scroll [41]	Native	~4 hours	Unique Identifier
Stargate [42]	Multichain	~4 mins	Token Transfer
Synapse [44]	Multichain	10-20 mins	Token Transfer
Wormhole [48]	Multichain	up 24 hours	Token Transfer
zkSync Era [51]	Native	~15 mins	Unique Identifier

Table 4: Overview of detected native and multichain bridges.

3.4 Validation

We first apply the approach defined in Section 3.1 to identify the set of 2-hop cross-chain arbitrage matches across the nine blockchain networks examined. However, these matches may still include false positives, as two unrelated addresses could coincidentally execute trades on the same tokens, in opposite directions, with similar amounts at close times. This makes cross-chain arbitrage detection more complex than atomic arbitrage, which is completed in a single transaction and requires no such further validation for accurate identification.

To address this challenge, we ensure that the entities behind the transactions are associated. We verify whether the sender addresses (i.e., Externally Owned Accounts (EOAs)) or the initially interacted smart contracts are identical. For contracts, we further check that they do not belong to a known application (e.g., a DEX or an aggregator). Identical EOAs across blockchains indicate access to the same private key, suggesting explicit intent to execute the arbitrage. If the contracts are the same but not associated with a known application, they are likely arbitrage-specific contracts that can only be used by their owners. If either of these two conditions is satisfied for a match, we classify it as a correctly identified cross-chain arbitrage. Otherwise, we filter it out.

3.5 Datasets

In the following, we describe the various datasets used to detect historical cross-chain arbitrages executed over a one-year period from September 2023 to August 2024.

3.5.1 Trading Data

We collected transaction-level trading data for our nine blockchains using Allium [4], an external blockchain data provider. All swap and aggregator events, along with their respective transaction data, were gathered, and transactions likely involved in atomic arbitrages were filtered out by ensuring their swaps did not form cycles. Detecting swap events across different blockchains is challenging due to the variety of DEXes and aggregators available, many of which use uncommon event signatures. Through Allium’s protocol labels, we identified 108 different protocols involved in the cross-chain arbitrages that we detected, which we believe provides sufficient coverage.

While standard DEX trades are straightforward to analyze, aggregator transactions are more complex, as they often involve multiple swaps within a single transaction. To address this, we developed logic to track state changes between input and output tokens, consolidating them into a single trade. Additionally, we identified generic asset symbols (e.g., stablecoins or native currency versions like bridged and wrapped tokens) to prevent valid cross-chain arbitrages from being excluded due to token mismatches (e.g., USDC vs. USDT).

For each trade transaction, we collected metadata such as transaction fees and coinbase payments (when applicable), which are necessary for evaluating arbitrage costs. Additionally, Allium provides USD conversions for trade volumes and transaction fees based on their hourly token pricing model [5]. In total, we processed over 530 million trade transactions executed across the nine different blockchains during a period of one year.

3.5.2 Bridge Interactions

Our bridge detection method based on token transfers might produce false positives as it relies on heuristics to match cross-chain bridge transactions. To eliminate potential false positives we leverage Allium’s dataset of labeled bridge transactions to filter out any identified cross-chain transactions which are not part of Allium’s dataset. Allium’s dataset also provides us with a label detailing what specific bridge was used by the bridge transaction, which we cannot get from solely looking at token transfer events.

3.5.3 Smart Contract Labels

To identify smart contract addresses potentially associated with cross-chain arbitrage bots, we curated a dataset of labeled smart contracts known not to be associated with bots, including DeFi applications and other non-MEV related contracts. We compiled unique protocol addresses from Allium and Dune Analytics [17] to ensure comprehensive coverage. Additionally, we incorporated the addresses of known non-MEV contracts on Ethereum [46] and market maker contracts we have identified. This process resulted in a dataset of 1,285,465 unique addresses, which we use to assess whether an interacted contract might belong to a cross-chain arbitrage bot.

3.5.4 Ethereum Mempool Data

We make use of the Mempool Dumpster [24] dataset by Flashbots to identify private Ethereum transactions. This dataset includes entries for Ethereum transactions observed by node providers in the mempool before being included in a block. Transactions missing from this dataset are likely privately relayed to Ethereum block builders via endpoints like Flashbots Protect [23] and MEV Blocker [32].

3.6 Limitations

Our methodology currently detects a lower-bound of cross-chain arbitrages, as it is restricted to 2-hop arbitrages and relies on specific heuristics (e.g., time and marginal difference thresholds, associated entities) to minimize false positives. While we cover a wide range of DeFi protocols on the studied blockchains, recently introduced protocols may be excluded if they are not tracked by Allium. Similarly, some bridge interactions may be missed for the same reason. Additionally,

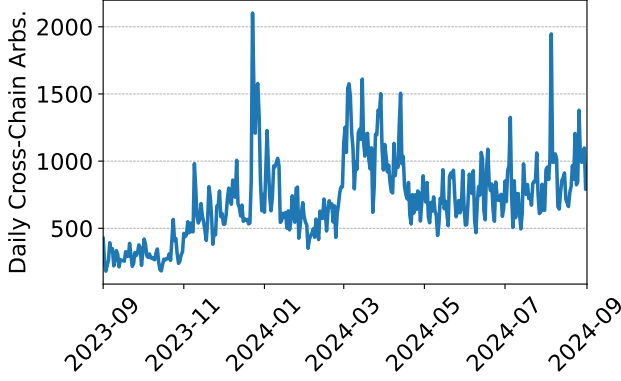


Figure 2: Daily number of cross-chain arbitrages executed between September 2023 and August 2024.

Allium’s pricing model may lack data for less popular tokens, preventing USD profit calculations for arbitrages involving these assets. Our profit calculation also does not account for inventory risks. Furthermore, our results may include false positives for token pairs available on CEXes, particularly those traded by market makers or used in CEX-DEX arbitrages. Finally, our Ethereum mempool visibility is limited by Mempool Dumpster’s network coverage.

4 Cross-Chain Arbitrage Analysis

In this section, we analyze the cross-chain arbitrages detected via our methodology, by providing an overview and then examining activity across different blockchain pairs, arbitrage durations, tokens involved, and bridge usage.

4.1 Overview

We detected 260,808 cross-chain arbitrages between September 2023 and August 2024, executed across the nine blockchains examined. Figure 2 illustrates the daily number of arbitrages executed. While certain periods show activity spikes, we observe an overall upward trend. Notably, the average daily number of arbitrages increased by 3.22 x from the first to the last month.

These cross-chain arbitrages cumulatively traded approximately 465,797,487.98 USD in volume, generating 10,983,325.55 USD in revenue for arbitrageurs. Notably, the arbitrageurs maintained an average profit margin of 66.92 %, yielding a total profit of 9,496,115.28 USD. In comparison, atomic arbitrages typically achieve profit margins below 10 % [13], highlighting the superior return rates of cross-chain arbitrages. We attribute this difference to the higher risks involved in cross-chain arbitrages, which require maintaining inventory across multiple networks or using bridging solutions. These factors introduce execution risks and raise

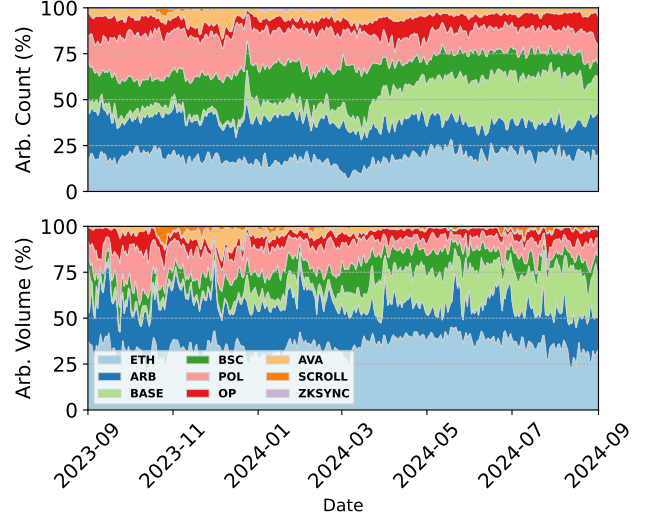


Figure 3: Daily percentage of cross-chain arbitrages executed by count (top) and USD volume (bottom) across nine blockchains.

barriers to entry, as they demand more capital to operate compared to single-domain MEV. In contrast, atomic arbitrages are risk-free, as they either succeed or fail entirely, leading to more intense competition since participation is more accessible.

[13] estimates a profit margin of 23 %–65 % for CEX-DEX arbitrageurs, which aligns closely with cross-chain arbitrages. This similarity likely stems from the shared non-atomic and risky nature of both strategies.

4.2 Blockchain Pairs

Across the nine blockchains that we examined, Arbitrum accounts for the highest percentage of cross-chain arbitrages, contributing 20.36 % of the total. However, despite being involved in fewer arbitrages (17.84 %), Ethereum is involved in 36 % of the total arbitrage trade volume. We suspect this is due to the significant liquidity available on Ethereum DEXes. The time-series in Figure 3 shows that Ethereum maintained a relatively consistent share in both count and volume over time, while blockchains like Base gained traction as time progressed.

Figure 4 presents heatmaps of cross-chain arbitrage activity between blockchains. While Arbitrum and Binance Smart Chain are heavily involved in arbitrages (see Figure 4a), Ethereum stands out in terms of arbitrage volume and profits (see Figure 4b and Figure 4c), consistent with the trends observed in Figure 3. This is further supported by the metrics summarized in Table 5 for the top blockchain pairs contributing ≥ 1 % of the total arbitrage volume. Notably, the top four highest-volume pairs all involve Ethereum and also generate the highest profits.

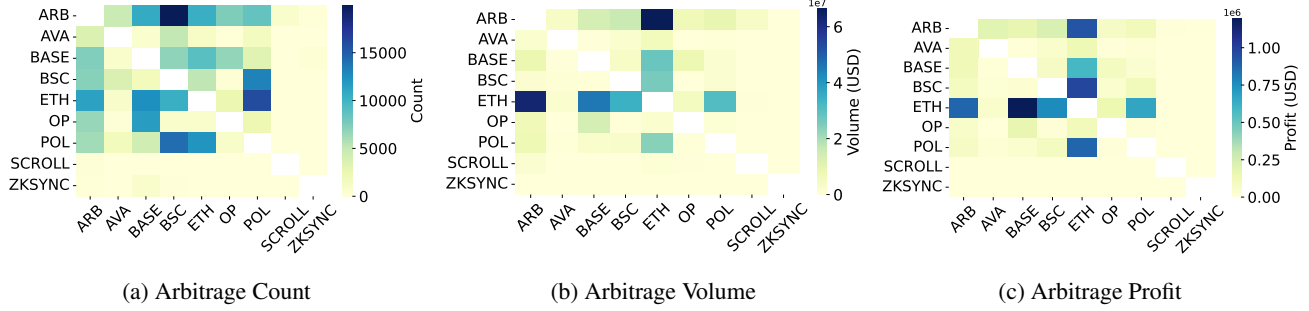


Figure 4: Heatmaps of cross-chain arbitrage between blockchains. The y-axis indicates the starting blockchain, and the x-axis represents the ending blockchain of an arbitrage, determined by transaction timestamps.

Interestingly, the second- and third-highest number of arbitrages occurred on ARB-BSC and BSC-POL pairs. However, these pairs exhibited relatively lower trading volumes, likely due to the low-liquidity token pairs utilized. In contrast, ARB-SCROLL arbitrages, though rare, conducted relatively high trading volumes. Further inspection reveals that 99.6 % of these arbitrages involved WETH-USDC and WETH-WBTC pairs, which are typically traded on CEXes. Similar patterns are observed across other blockchain pairs involving Scroll (see Table 9 in Appendix A).

4.3 Arbitrage Durations

We analyze the time difference between the two legs of cross-chain arbitrages to infer their durations. Figure 5 reveals a positively skewed distribution, followed by a long tail of arbitrage durations capped at one hour by our methodology (see Section 3.1). The spike observed around 1,200 s–1,500 s is attributed to arbitrages between Ethereum and Polygon, the most frequently arbitrated blockchain pair, with a median

duration of 1,247 seconds. This is potentially due to the extensive bridging durations involved (see Table 4). For detailed metrics for on arbitrage durations across all blockchain pairs, refer to Table 9 in Appendix A.

Notably, 34 % of all arbitrages are completed in under 10 seconds. However, the existence of arbitrages with longer durations, compared to the time-sensitive nature of CEX-DEX arbitrages, where prices on CEXes update within milliseconds, suggests that competition for cross-chain opportunities is less intense. This could be attributed to the token pairs involved, which may remain stale for extended periods, allowing arbitrage positions to stay open.

4.4 Involved Tokens

We examine the most frequently used and profitable token pairs. Our analysis reveals that OmniCat (OMNI), an interoperable token across blockchains, is the most arbitrated token, accounting for 4.84 % of total arbitrages. Overall, we detect 1,944 distinct token pairs involved in cross-chain arbitrages, with a Herfindahl–Hirschman Index (HHI) [38] of 0.007, indicating minimal concentration among arbitrated pairs.

Chain Pair	Total Volume [USD]	Total Profit [USD]	Total Arbs. [#]	Avg. Volume [USD]	Avg. Profit [USD]
ARB-ETH	129.3 M	1.8 M	21,888	5,959.77	83.42
BASE-ETH	73.2 M	1.8 M	21,466	3,449.82	84.60
BSC-ETH	60.0 M	1.7 M	15,763	3,881.62	112.98
ETH-POL	54.8 M	1.5 M	28,247	1,944.98	54.52
ARB-BASE	22.6 M	300.4 K	18,245	1,261.83	16.92
BASE-OP	22.1 M	266.0 K	18,559	1,228.07	15.16
ARB-BSC	17.7 M	318.5 K	27,182	655.01	11.83
ARB-POL	16.7 M	193.2 K	14,823	1,134.84	13.18
ARB-OP	14.0 M	141.6 K	14,138	1,044.48	10.85
AVA-ETH	8.4 M	187.9 K	2,100	4,020.15	92.91
ETH-OP	7.7 M	263.1 K	3,459	2,269.92	77.74
ARB-AVA	6.4 M	317.0 K	8,430	772.45	40.25
ARB-SCROLL	6.1 M	8.5 K	1,117	5,436.93	7.62
BSC-POL	5.5 M	211.1 K	27,468	202.20	7.86
BASE-POL	5.0 M	84.4 K	7,737	655.57	11.01

Table 5: Cross-chain arbitrage metrics for top blockchain pairs contributing $\geq 1\%$ of the total arbitrage volume.

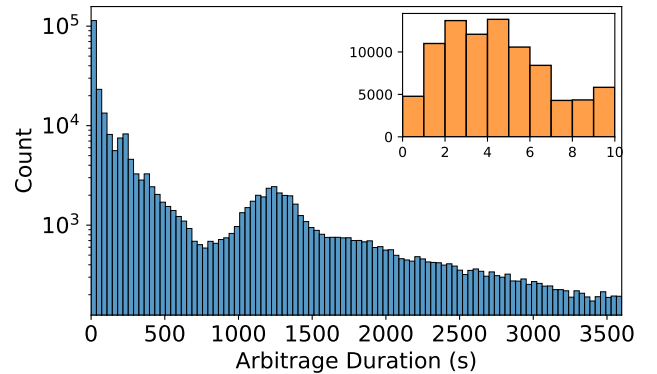


Figure 5: Distribution of cross-chain arbitrage durations with an inset plot highlighting the first ten seconds where arbitrages frequently occur.

Unlike CEX-DEX arbitrages, which are dominated by highly liquid and actively traded pairs such as WETH-USDC, cross-chain arbitrages often involve pairs with limited liquidity and low trading frequency. Our analysis reveals that half of the top ten most profitable tokens have a Fully Diluted Valuation (FDV) of less than 65 M USD (see Table 6), which is negligible compared to tokens like WETH, USDC, or WBTC, each with FDVs in the billions [14]. Additionally, activity on these most profitable tokens typically exhibits sharp spikes during specific periods, with minimal usage at other times (see Figure 6).

4.5 Bridge Usage

We find that 67.63 % of cross-chain arbitrages *do not* use a bridge, indicating that the SIA strategy is significantly more prevalent. This discrepancy in adoption between SIA and SDA strategies may be attributed to the time-sensitive nature of arbitrages, where price deviations can quickly render opportunities unprofitable. Arbitrageurs avoiding bridges, which introduce additional delays and costs, are thus at an advantage. Our analysis reveals that SIAs have a median completion time of 10 seconds, compared to 246 seconds for SDAs. By eliminating the need to bridge assets, the SIA strategy saves critical time, which is often essential for successful execution.

Interestingly, we discover that SIA arbitrages yield a lower average profit compared to SDA arbitrages, with 34.8 USD versus 43.5 USD, respectively. While this result may seem counterintuitive, it could stem from limitations in our profit calculation methodology (see Section 3.6) or market making activity being misclassified as cross-chain arbitrages.

Among SDAs, we observe that only 26.68 % utilize native bridges, while the majority rely on multichain solutions. Notably, nearly half of the SDAs employ the LayerZero bridge, aligning with tokens frequently involved in cross-chain arbitrages (see Table 6), such as OMNI, which depends on LayerZero for cross-blockchain transfers.

We also identify significant differences in average bridging durations and costs across bridging protocols, as summarized

Token Pair	Total Profit [USD]	Total Volume [USD]	Total Arbs. [%]	Avg. Profit [USD]
NMT-USDC	645,398.53	28,002,522.18	0.72 %	344.58
OMNI-WETH	455,331.04	17,091,969.10	4.84 %	36.60
DMT-WETH	339,690.54	24,212,698.85	1.52 %	85.78
DOG-WETH	237,422.19	5,795,675.49	0.31 %	297.52
USDC-agEUR	220,401.71	10,747,540.46	1.34 %	72.43
TEL-WETH	191,799.36	4,865,794.60	0.76 %	96.77
Mog-WETH	176,209.68	7,526,070.06	0.53 %	126.86
PRIME-WETH	165,903.80	15,070,549.59	0.60 %	105.81
GENE-WETH	158,834.98	4,169,789.04	0.87 %	70.69
NPC-WETH	157,382.60	9,787,120.85	1.45 %	41.56

Table 6: Statistics on the top ten most profitable token pairs.

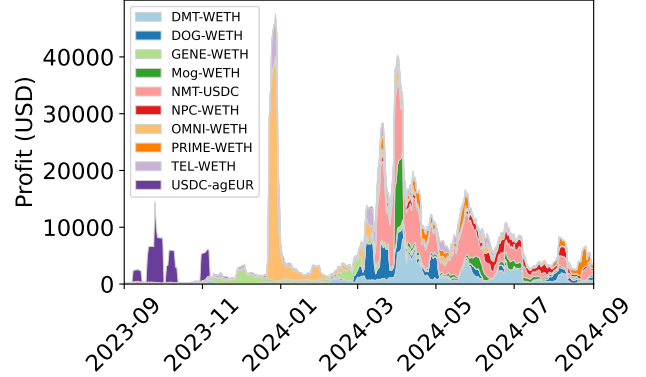


Figure 6: Daily profits generated by the top ten most profitable token pairs.

in Table 7. Native bridge solutions offered by blockchains are among the slowest and charge the highest fees, accounting for 30.27 % of the total fees paid by a cross-chain arbitrage. This highlights a potential trade-off between bridge security and arbitrage speed. However, we observe that native bridge arbitrages yield nearly double the average profit of multichain arbitrages, with 60.39 USD compared to 37.02 USD. This suggests that arbitrageurs could be trusting native bridges for higher-volume arbitrages, while multichain solutions are preferred for arbitrages requiring faster execution.

Figure 7 shows that while the execution of the SIA strategy has increased significantly over time, the growth in multichain and native bridge arbitrages has been comparatively modest. This trend raises questions about the efficiency and usability of current bridging technologies for cross-chain arbitrages, as SDA adoption remains limited.

5 Security Implications

In this section, we examine the security implications of the detected cross-chain arbitrages, focusing on arbitrageur centralization, network congestions, and private mempool usage.

Bridge	Total Arbs. [#]	Avg. Time [s]	Avg. Cost [USD]	Avg. Cost [%]
LayerZero	42,537	179.73	1.65	26.87
Native	22,529	729.05	4.99	30.28
Wormhole	8,801	998.28	2.04	29.56
Axelar	3,239	1367.65	0.45	21.52
Celer	3,129	315.78	0.72	24.98
Across	2,849	285.02	0.29	19.13
Synapse	823	296.49	1.22	22.04
Stargate	367	192.49	0.88	26.47

Table 7: Summary statistics for detected bridge protocols.

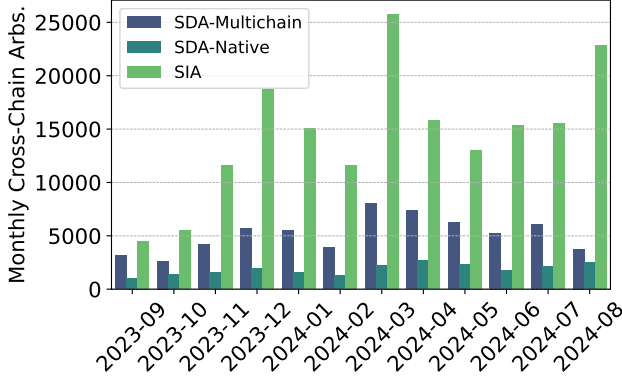


Figure 7: Monthly number of cross-chain arbitrages by strategy types.

5.1 Arbitrageur Centralization

MEV has been shown to exert significant centralization pressure across various layers of the blockchain ecosystem. This centralization is primarily driven by economies of scale, leading to the concentration of entities extracting value. In the context of cross-chain MEV, this dynamic could result in a privileged group of “super-traders” [33] controlling MEV extraction across multiple blockchains. By optimizing strategies tailored to specific network properties, such as transaction sequencing rules, these entities further increase barriers to entry for new players [33].

Our analysis reveals that a single arbitrageur, `0xCA74`, is responsible for 32.79 % of all cross-chain arbitrage volume, 18.06 % of all arbitrages, and 23.07 % of total profits. In particular, `0xCA74`’s dominance has grown over time, as shown in Figure 8, with 43.15 % of the total volume executed in the last three months. Although other arbitrageurs showed significant activity during certain periods, none matches the consistent and increasing dominance of `0xCA74`.

Table 10 in Appendix B summarizes further metrics for the arbitrageurs with the highest trading volume. We observe that SIA strategy is commonly adopted, which could act as a centralization vector since it requires active inventory management across multiple blockchains. Similarly, the way arbitrages are executed could be another aspect creating centralization pressure. We find that most arbitrageurs are not using smart contracts to execute their strategies but rather EOAs. In atomic arbitrages, deploying a smart contract offers significant advantages, such as implementing logic to identify profitable opportunities and ensuring execution only when conditions are met, thereby eliminating risk. Additionally, solutions offering revert protection [23, 32] make this approach even more accessible. However, in cross-chain arbitrages, this is not feasible because there is currently no straightforward way for smart contracts deployed on different networks to synchronize and guarantee successful atomic execution. Thus,

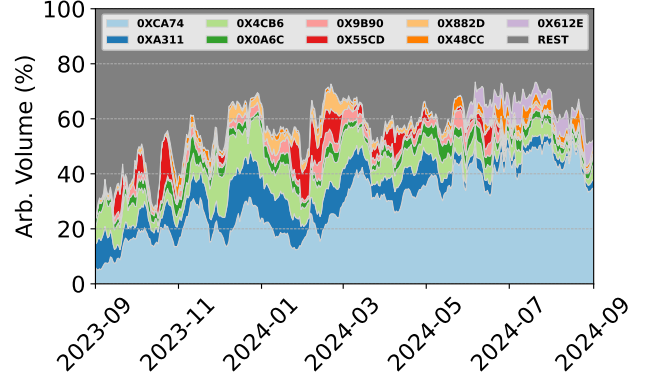


Figure 8: Daily percentage of cross-chain arbitrage volume (USD) executed by the top arbitrageurs by trade volume compared to the rest of the arbitrageurs over time.

arbitrageurs must implement off-chain services to process signals across multiple blockchains and determine when and where to issue transactions with the optimal input parameters.

5.2 Network Congestion

Network congestion has been identified as a negative side effect of MEV extraction competition on blockchains, often filling blocks with reverted transactions as a result of failed attempts to extract value [15, 22]. These reverted transactions have no effect on the global blockchain state, yet they are included into the block, thereby occupying block space and potentially forcing other transactions out of the current block or even increasing transaction fees for other users.

In the context of cross-chain MEV, this impact can be amplified, as competition for an opportunity spans multiple blockchains. However, detecting reverted cross-chain arbitrages is particularly challenging, as it requires associating reverted transactions across blockchains. To address this, we adopt a more pragmatic approach by examining all reverted transactions generated by cross-chain arbitrageurs, without distinguishing their specific purpose. While this method does not directly measure network congestion caused solely by cross-chain arbitrage competition, we believe the results for the most successful arbitrageurs provide relevant insights. To limit false positives, we focus only on top arbitrageurs identified via their smart contracts, as outlined in Table 10 in Appendix B.

Table 8 reveal significantly high failure rates among the top four arbitrageurs identified as using smart contracts. Notably, `0x882D` and `0x826A` generate nearly nine failed transactions for every successful one. Interestingly, Polygon emerges as the blockchain with the highest number of failures for three of these arbitrageurs. Overall, these high revert rates may be attributed to the absence of private endpoints offering revert protection, such as those available on Ethereum [23, 32].

Arbitrageur	Total Success [#]	Total Fail [#]	Fail Rate [%]	Fail Distribution [#]
0X882D	5,472	72,884	93.02	POL: 70,422, ETH: 2,462
0X826A	4,420	65,810	93.71	POL: 64,564, ETH: 1,246
0X0A6C	11,288	24,111	68.11	POL: 12,729, BSC: 6,286
0XA311	12,745	8,103	38.87	ARB: 3,235, BSC: 1,473

Table 8: Reverted transaction metrics for the top cross-chain arbitrageurs using smart contracts. The last column indicates the top two blockchains where most failures occurred.

5.3 Private Mempools

The competition on Ethereum among MEV extractors has led to the emergence of private mempools, which protect transactions from being exposed in the public mempool and getting frontrun [23, 32]. This is critical, as the successful execution of an MEV strategy often depends on acting on a specific state before other users. To investigate this phenomenon, we focus on cross-chain arbitrages involving Ethereum.

Our analysis reveals that 32.88 % of Ethereum legs in cross-chain arbitrages were submitted privately, yielding an average profit of 82.5 USD, compared to 78.35 USD for public transactions. Although the profit difference is not substantial, the results align with our intuition that private mempool usage can be advantageous for cross-chain arbitrageurs. Additionally, the monthly number of privately submitted transactions increased by 5.52 x from September 2023 to August 2024, indicating an overall upward trend (see Figure 9).

When comparing the positioning of public and private Ethereum cross-chain arbitrages, we find that private transactions are more frequently placed in top block positions, giving arbitrageurs an advantage by prior execution. Specifically, private transactions have a normalized median position index of 7.11 %, lower than the 23.06 % observed for public transactions. A detailed distribution is illustrated in Figure 10. Further analysis of transaction positioning across other blockchains is provided in Appendix C.

6 Discussion

In this section, we discuss how cross-chain arbitrages can incentivize controlling sequencing across multiple blockchains and propose a risk-optimized arbitrage strategy.

6.1 Sequencer Incentives

The risky nature of cross-chain arbitrages is evidenced by the long durations observed for certain blockchain pairs (e.g., Ethereum and Polygon - see Table 9), and the high failure rates of arbitrageurs utilizing their own smart contracts (see Table 8). To mitigate these risks, controlling transaction sequencing in blocks can be crucial. However, as cross-chain

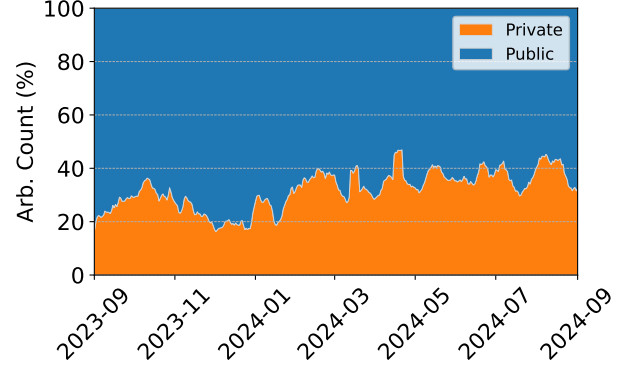


Figure 9: Percentage of cross-chain arbitrage transactions submitted privately over time on Ethereum.

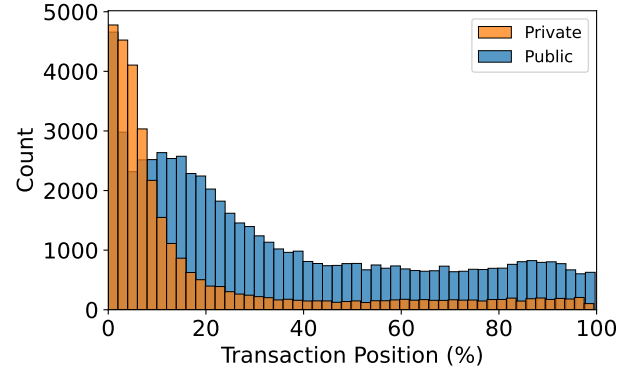


Figure 10: Distribution of normalized cross-chain arbitrage transaction positions (%) for public and private transactions on Ethereum.

arbitrages gain prominence (see Figure 2), controlling sequencing rights across multiple blockchains is anticipated to pose a significant centralization threat [33]. Entities with established infrastructure and staked validators across multiple blockchains could be incentivized to collude [33], enabling them to execute cross-chain arbitrages atomically and without risk. To prevent such centralization risk undermining economic security and ensuring fair access to cross-chain MEV, shared-sequencing solutions like Espresso Systems [19] and Astria [8] could play an important role. These solutions can lower the barriers to entry for cross-chain arbitrageurs by offering sequencing services on multiple networks and providing execution guarantees within the cross-chain ecosystem.

6.2 Risk-Optimized Strategy

Arbitrageurs who lack inventory across multiple blockchains are forced to use bridges to execute their strategies. To overcome this limitation, we propose a risk-optimized strategy that leverages lending protocols to complete arbitrages similarly to an SIA, while still using bridging to close the loan

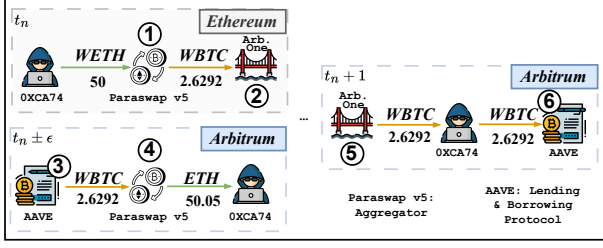


Figure 11: Example of a risk-optimized cross-chain arbitrage strategy between Ethereum and Arbitrum, where the arbitrageur completes the arbitrage at $t_n \pm \epsilon$ using a loan and later closes the loan position with the bridged output of the hedge leg upon its arrival at $t_n + 1$.

position after the arbitrage is executed. Figure 11 illustrates a real-world example of such a risk-optimized arbitrage, executed between Ethereum and Arbitrum by the arbitrageur 0xCA74. The process unfolds as follows:

1. 0xCA74 initially swaps 50 WETH for 2.629 WBTC on Ethereum (0xf9d...08a).
2. The WBTC is bridged to Arbitrum in the next block (0x452...903).
3. 8 seconds later, 0xCA74 takes a loan for 2.629 WBTC using Aave [2] (0xf7e...eff).
4. The WBTC is then swapped back to ETH, earning 0.5 ETH and completing the arbitrage (0xdd4...72f).
5. The bridged WBTC arrives on Arbitrum around 7 minutes after the arbitrage is completed (0x701...1bf).
6. Finally, 0xCA74 transfers the bridged WBTC to AAVE to close the loan, concluding the strategy (0x798...935).

7 Related Work

Qin et al. [37] provided the first comprehensive quantification of MEV, documenting systematic value extraction through sandwich attacks, liquidations, and DEX arbitrage on Ethereum. Torres et al. [21] on the other hand presented the first empirical quantification of frontrunning strategies such as displacement, insertion, and suppression, thereby building up on the theoretical work presented by Eskandari et al. [18]. McLaughlin et al. [30] advanced this understanding by developing an application-agnostic arbitrage detection methodology using standardized ERC-20 transfer events, enabling analysis across a significantly broader range of decentralized exchanges. Torres et al. [22] extended MEV research into L2 solutions by investigating MEV activities across Arbitrum, Optimism, and zkSync, revealing comparable trading volumes to Ethereum on rollups, albeit with significantly lower profits despite reduced costs. The analysis of non-atomic arbitrage has been advanced by Heimbach et al. [26], who demonstrated

that around 25 % of volume on Ethereum’s top five DEXes could be attributed to arbitrage between CEXes and DEXes, revealing substantial centralization among a small group of dominant MEV searchers. Obadia et al. [33] provided the first formal definition of cross-domain MEV by introducing domains as self-contained systems with shared states and showing how cross-domain MEV opportunities incentivize sequencer collusion across chains. McMenamin [31] systematized cross-domain MEV by categorizing extractable value based on extraction methods and value origins while analyzing different protocols’ MEV mitigation capabilities. Gogol et al. [25] provided initial insights into cross-rollup arbitrage by analyzing price disparities in WETH-USDC trading pairs between Ethereum and major L2 rollups, showing that arbitrage opportunities persisted for multiple blocks with relatively limited profits ranging from 0.03 % to 0.25 % of trading volume. Mazorra et al. [29] provided theoretical foundations by developing an abstract framework for analyzing MEV games across domains, particularly focusing on network characteristics’ influence on strategies and congestion. Finally, Öz et al.’s [36] analysis on block building revealed how exclusive order flow access by select builders leads to market centralization in Ethereum’s block building, highlighting similar centralization risks that could emerge from privileged access to cross-chain MEV opportunities.

8 Conclusion

MEV extraction is not confined to individual blockchains. As we have empirically demonstrated in this work, cross-chain arbitrages are already a reality. We presented two distinct strategies, SIA and SDA, employed by arbitrageurs to perform cross-chain arbitrages in practice. Our findings reveal that the majority of these arbitrages adopt the SIA strategy, which avoids the delays and costs associated with bridging. Additionally, we highlighted significant centralization among arbitrageurs, with a single entity dominating the space. We also observed high failure rates among cross-chain arbitrageurs, potentially contributing to network congestion, and noted the growing adoption of private mempools for transaction privacy and prioritization. Finally, we discussed how the increasing prominence of cross-chain arbitrages could incentivize seeking control over transaction sequencing across multiple blockchains, emphasizing the importance of shared-sequencing solutions to enable a level playing field.

Acknowledgements

This work was financially supported by the Composable Finance Foundation under project references UIDB/50021/2020 and 1601681. We extend our gratitude to Danut Ilisei, Murad Muradli, and Thomas Wagner for their contributions. We also thank Quintus Kilbourn for his valuable feedback.

References

- [1] 48 CLUB DOCS. Send bundle. URL: <https://docs.48.club/puissant-builder/send-bundle>.
- [2] AAVE. Aave documentation. URL: <https://aave.com/docs>.
- [3] ACROSS DOCS. URL: <https://docs.across.to/user-docs/additional-info/faq>.
- [4] ALLIUM. URL: <https://www.allium.so/>.
- [5] ALLIUM. Prices, 2024. URL: <https://docs.allium.so/historical-data/prices>.
- [6] ARBITRUM DOCS. URL: <https://docs.arbitrum.io/arbitrum-bridge/quickstart>.
- [7] ARBITRUM DOCS. A gentle introduction: Timeboost. URL: <https://docs.arbitrum.io/how-arbitrum-works/timeboost/gentle-introduction>.
- [8] ASTRIA. URL: <https://www.astria.org/>.
- [9] AXELAR BLOG. URL: <https://www.axelar.network/blog/arbitrum-bridge>.
- [10] BINANCE. URL: <https://www.binance.com>.
- [11] BNB CHAIN. URL: <https://docs.bnbchain.org>.
- [12] CELER CBRIDGE. URL: <https://cbridge-docs.celer.network/reference/faq>.
- [13] CHEN, E., TOBEROFF, A., SRINIVASAN, S., AND CHIPLUNKAR, A. A tale of two arbitrages. URL: <https://frontier.tech/a-tale-of-two-arbitrages>.
- [14] COINMARKETCAP. URL: <https://coinmarketcap.com/>.
- [15] DAIAN, P., GOLDFEDER, S., KELL, T., LI, Y., ZHAO, X., BENTOV, I., BREIDENBACH, L., AND JUELS, A. Flash boys 2.0: Frontrunning in decentralized exchanges, miner extractable value, and consensus instability. In *2020 IEEE Symposium on Security and Privacy (SP)* (2020), pp. 910–927.
- [16] DEFILLAMA. URL: <https://defillama.com/>.
- [17] DUNE ANALYTICS. URL: <https://dune.com/home>.
- [18] ESKANDARI, S., MOOSAVI, S., AND CLARK, J. Sok: Transparent dishonesty: Front-running attacks on blockchain. In *Financial Cryptography and Data Security - FC 2019 International Workshops, VOTING and WTSC, St. Kitts, St. Kitts and Nevis, February 18-22, 2019, Revised Selected Papers* (2019), A. Bracciali, J. Clark, F. Pintore, P. B. Rønne, and M. Sala, Eds., vol. 11599 of *Lecture Notes in Computer Science*, Springer, pp. 170–189.
- [19] ESPRESSO SYSTEMS. URL: <https://www.espressosys.com/>.
- [20] FASTLANE LABS. What is polygon fastlane? URL: <https://fastlane-labs.gitbook.io/polygon-fastlane>.
- [21] FERREIRA TORRES, C., CAMINO, R., AND STATE, R. Frontrunner jones and the raiders of the dark forest: An empirical study of frontrunning on the ethereum blockchain. In *30th USENIX Security Symposium, USENIX Security 2021, August 11-13, 2021* (2021), M. D. Bailey and R. Greenstadt, Eds., USENIX Association, pp. 1343–1359.
- [22] FERREIRA TORRES, C., MAMUTI, A., WEINTRAUB, B., NITAROTARU, C., AND SHINDE, S. Rolling in the shadows: Analyzing the extraction of mev across layer-2 rollups. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security* (New York, NY, USA, 2024), CCS ’24, Association for Computing Machinery, p. 2591–2605.
- [23] FLASHBOTS. Flashbots protect. URL: <https://docs.flashbots.net/flashbots-protect/overview>.
- [24] FLASHBOTS. Mempool dumpster. URL: <https://github.com/flashbots/mempool-dumpster>.
- [25] GOGOL, K., MESSIAS, J., MIORI, D., TESSONE, C., AND LIVSHITS, B. Cross-Rollup MEV: Non-Atomic Arbitrage Across L2 Blockchains, Oct. 2024. arXiv:2406.02172.
- [26] HEIMBACH, L., PAHARI, V., AND SCHERTENLEIB, E. Non-atomic arbitrage in decentralized finance. In *2024 IEEE Symposium on Security and Privacy (SP)* (Los Alamitos, CA, USA, may 2024), IEEE Computer Society, pp. 224–224.
- [27] HILDOBBY. Atomic arbitrages. URL: <https://dune.com/hildobby/atomic-mev>.
- [28] LABS, P. URL: <https://docs.polygon.technology/pos/how-to/bridging/ethereum-polygon/matic-to-ethereum>.
- [29] MAZORRA, B., REYNOLDS, M., AND DAZA, V. Price of mev: Towards a game theoretical approach to mev. *Proceedings of the 2022 ACM CCS Workshop on Decentralized Finance and Security* (2022).
- [30] MCLAUGHLIN, R., KRUEGEL, C., AND VIGNA, G. A large scale study of the ethereum arbitrage ecosystem. In *32nd USENIX Security Symposium (USENIX Security 23)* (Anaheim, CA, Aug. 2023), USENIX Association, pp. 3295–3312.
- [31] MCMENAMIN, C. Sok: Cross-domain mev, 2023.
- [32] MEV BLOCKER. URL: <https://cow.fi/mev-blocker>.
- [33] OBADIA, A., SALLES, A., SANKAR, L., CHITRA, T., CHELLANI, V., AND DAIAN, P. Unity is strength: A formalization of cross-domain maximal extractable value, 2021.
- [34] OPTIMISM DOCS. URL: <https://docs.optimism.io/builders/app-developers/bridging/standard-bridge>.
- [35] ÖZ, B., KRANER, B., VALLARANO, N., KRUGER, B. S., MATTHES, F., AND TESSONE, C. J. Time moves faster when there is nothing you anticipate: The role of time in mev rewards. In *Proceedings of the 2023 Workshop on Decentralized Finance and Security* (New York, NY, USA, 2023), DeFi ’23, Association for Computing Machinery, p. 1–8.
- [36] ÖZ, B., SUI, D., THIERY, T., AND MATTHES, F. Who Wins Ethereum Block Building Auctions and Why? In *6th Conference on Advances in Financial Technologies (AFT 2024)* (Dagstuhl, Germany, 2024), R. Böhme and L. Kiffer, Eds., vol. 316 of *Leibniz International Proceedings in Informatics (LIPIcs)*, Schloss Dagstuhl – Leibniz-Zentrum für Informatik, pp. 22:1–22:25.
- [37] QIN, K., ZHOU, L., AND GERVAIS, A. Quantifying blockchain extractable value: How dark is the forest? In *2022 IEEE Symposium on Security and Privacy (SP)* (2022), IEEE, pp. 198–214.
- [38] RHOADES, S. A. The herfindahl-hirschman index. *Federal Reserve Bulletin* (1993), 188–189.
- [39] ROCKET, T., YIN, M., SEKNIQI, K., VAN RENESSE, R., AND SIRER, E. G. Scalable and probabilistic leaderless bft consensus through metastability, 2020.
- [40] SCHWARZ-SCHILLING, C., SALEH, F., THIERY, T., PAN, J., SHAH, N., AND MONNOT, B. Time Is Money: Strategic Timing Games in Proof-Of-Stake Protocols. In *5th Conference on Advances in Financial Technologies (AFT 2023)* (Dagstuhl, Germany, 2023), J. Bonneau and S. M. Weinberg, Eds., vol. 282 of *Leibniz International Proceedings in Informatics (LIPIcs)*, Schloss Dagstuhl – Leibniz-Zentrum für Informatik, pp. 30:1–30:17.
- [41] SCROLL DOCUMENTATION. URL: <https://docs.scroll.io/en/user-guide/bridge>.
- [42] SMART CONTRACT AUDITS - COMPOSABLE SECURITY. URL: <https://composable-security.com/blog/secure-integration-with-layer-zero>.
- [43] SUPERBRIDGE DOCUMENTATION. URL: <https://docs.superbridge.app/what-is-bridging>.
- [44] SYNAPSE DOCS. URL: <https://docs.synapseprotocol.com/docs/Support/Transaction-Support>.

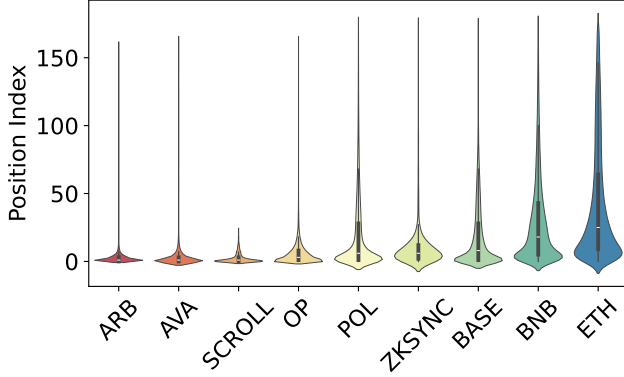


Figure 12: Distribution of cross-chain arbitrage position indexes per blockchain. Only the first 99th percentile is included.

- [45] UNISWAP. URL: <https://app.uniswap.org>.
- [46] WINNSTERX. searcherbuilder.pics. URL: https://github.com/winnsterx/searcherbuilder.pics/blob/main/labels/non_mev_contracts.py.
- [47] WOOD, G., ET AL. Ethereum: A secure decentralised generalised transaction ledger. *Ethereum project yellow paper 151*, 2014 (2014), 1–32.
- [48] WORMHOLE DOCS. URL: <https://wormhole.com/docs/build/contract-integrations/native-token-transfers/configuration/rate-limiting>.
- [49] YANG, S., NAYAK, K., AND ZHANG, F. Decentralization of ethereum’s builder market, 2024.
- [50] YANG, S., ZHANG, F., HUANG, K., CHEN, X., YANG, Y., AND ZHU, F. Sok: Mev countermeasures. In *Proceedings of the Workshop on Decentralized Finance and Security* (New York, NY, USA, 2024), DeFi ’24, Association for Computing Machinery, p. 21–30.
- [51] ZKSYNC BRIDGE. URL: <https://portal.zksync.io/bridge>.

Appendix

A All Blockchain Pairs

Table 9 summarizes the cross-chain arbitrage metrics for all blockchain pairs identified.

B Arbitrageurs

Table 10 summarizes the cross-chain arbitrage metrics for arbitrageurs with $\geq 1\%$ of the total trade volume.

C Arbitrage Transaction Positions

Transaction positioning plays a critical role in the successful execution of an MEV strategy. A concentration of transactions around lower position indexes indicates that arbitrageurs can effectively prioritize their transactions, ensuring execution at top block positions to capture opportunities first.

Figure 12 presents a violin plot showing the distribution of position indexes for cross-chain arbitrage transactions across the nine studied blockchains. Overall, we observe the highest density around smaller position indexes. However, blockchains like Binance Smart Chain and Ethereum exhibit higher median values and longer tails. This aligns with expectations, as these blockchains provide bundling solutions [1,23] that guarantee successful execution, though not necessarily at top block positions, given the prior execution guarantee.

While Arbitrum and Polygon offer transaction prioritization mechanisms such as Timeboost [7] and FastLane [20], respectively, we found no on-chain evidence suggesting significant adoption of these solutions for executing cross-chain arbitrages.

Chain Pair	Total			Strategy			Averages		Duration (Percentiles)		
	Volume [USD]	Profit [USD]	Arbs. [#]	SIA [%]	Multichain [%]	Native [%]	Volume [USD]	Profit [USD]	25th [s]	50th [s]	75th [s]
ARB-ETH	129,279,422.40	1,797,112.08	21,888	51.17	28.98	19.85	5,959.77	83.42	13.00	108.00	381.00
BASE-ETH	73,239,640.16	1,781,073.40	21,466	55.27	20.60	24.13	3,449.82	84.60	16.00	194.00	362.00
BSC-ETH	59,986,532.41	1,738,099.72	15,763	87.75	12.25	0.00	3,881.62	112.98	11.00	121.00	305.00
ETH-POL	54,815,332.43	1,533,726.03	28,247	58.15	2.95	38.91	1,944.98	54.48	160.50	1,247.00	1,882.00
ARB-BASE	22,550,090.67	300,390.57	18,245	69.09	30.91	0.00	1,261.83	16.92	2.00	31.00	172.00
BASE-OP	22,107,636.49	266,018.57	18,559	58.17	41.81	0.02	1,228.07	15.16	16.00	68.00	296.00
ARB-BSC	17,748,873.31	318,498.02	27,182	65.73	34.27	0.00	655.01	11.83	2.00	10.00	84.00
ARB-POL	16,692,353.34	193,239.27	14,823	82.22	17.76	0.01	1,134.84	13.18	5.00	9.00	277.00
ARB-OP	14,028,350.44	141,565.74	14,138	61.15	38.83	0.02	1,044.48	10.85	24.00	108.00	481.75
AVA-ETH	8,442,313.00	187,948.40	2,100	82.43	17.57	0.00	4,020.15	92.91	9.00	27.00	267.00
ETH-OP	7,726,790.92	263,073.91	3,459	34.69	8.73	56.58	2,269.92	77.74	78.00	100.00	158.00
ARB-AVA	6,436,022.80	316,962.95	8,430	40.06	59.94	0.00	772.45	40.25	14.00	39.00	103.00
ARB-SCROLL	6,073,053.19	8,508.52	1,117	100.00	0.00	0.00	5,436.93	7.62	3.00	5.00	18.00
BSC-POL	5,498,395.88	211,146.03	27,468	93.84	6.16	0.00	202.20	7.86	4.00	6.00	73.00
BASE-POL	5,043,298.38	84,425.77	7,737	80.83	18.95	0.22	655.57	11.01	6.00	133.00	470.00
BASE-BSC	4,452,286.73	108,430.30	8,791	72.90	27.10	0.00	509.65	12.79	3.00	55.00	262.50
AVA-BSC	3,607,246.38	80,944.07	8,885	86.28	13.72	0.00	409.54	9.22	3.00	6.00	60.00
OP-POL	1,983,611.41	22,956.48	3,370	47.74	51.69	0.56	604.02	7.09	179.00	399.50	957.50
ETH-SCROLL	1,839,985.92	3,469.43	119	69.75	26.05	4.20	17,036.91	32.12	21.00	29.00	501.00
AVA-POL	1,145,067.22	44,013.79	3,455	85.04	14.96	0.00	333.55	13.14	7.00	41.00	324.00
BSC-OP	843,907.46	57,155.77	1,275	60.00	40.00	0.00	671.90	46.77	7.00	121.00	760.00
AVA-BASE	587,816.71	24,582.11	1,638	60.38	39.62	0.00	361.51	15.84	6.00	48.00	127.00
OP-SCROLL	583,576.23	1,450.84	110	100.00	0.00	0.00	5,305.24	13.43	3.00	8.00	20.75
AVA-OP	318,318.56	8,447.80	454	40.09	59.91	0.00	774.50	24.56	61.00	190.00	727.75
POL-SCROLL	205,829.33	1,072.47	44	100.00	0.00	0.00	4,677.94	24.37	15.75	26.00	283.50
ARB-ZKSYNC	196,420.57	0.00	426	27.23	72.77	0.00	644.00	0.00	112.25	214.50	412.00
BASE-ZKSYNC	89,280.10	0.00	1,294	29.75	70.25	0.00	129.96	0.00	171.25	295.00	596.50
ETH-ZKSYNC	72,425.74	0.00	28	32.14	46.43	21.43	3,148.95	0.00	302.75	476.00	1,111.50
BASE-SCROLL	60,654.92	249.54	49	79.59	20.41	0.00	1,318.59	5.94	20.00	28.00	109.00
BSC-ZKSYNC	56,499.12	0.00	125	41.60	58.40	0.00	627.77	0.00	108.00	180.00	524.00
BSC-SCROLL	46,535.01	1,553.73	100	83.00	17.00	0.00	465.35	15.69	130.00	184.50	269.00
OP-ZKSYNC	26,971.82	0.00	18	66.67	33.33	0.00	3,371.48	0.00	13.75	19.50	208.00
SCROLL-ZKSYNC	12,945.96	0.00	4	100.00	0.00	0.00	4,315.32	0.00	11.50	12.00	14.75
POL-ZKSYNC	2.96	0.00	1	100.00	0.00	0.00	2.96	0.00	560.00	560.00	560.00

Table 9: Cross-chain arbitrage metrics for all blockchain pairs.

Arbitrageur		Total			Strategy			Averages		Duration (Percentiles)		
Address	Type	Volume [USD]	Profit [USD]	Arbs [#]	SIA [%]	Multichain [%]	Native [%]	Volume [USD]	Profit [USD]	25th [s]	50th [s]	75th [s]
0XCA74	EOA	152,739,321.26	2,191,157.66	47,119	68.15	18.55	13.30	3,248.18	46.70	6.00	68.00	580.00
0XA311 [†]	SC	33,480,140.90	590,530.94	12,745	49.81	47.58	2.61	2,631.67	46.74	3.00	27.00	58.00
0X4CB6	EOA	32,572,100.26	432,060.16	22,692	73.40	26.51	0.09	1,446.56	19.40	2.00	6.00	41.00
0X0A6C [†]	SC	14,510,776.67	288,433.32	11,288	21.77	34.95	43.28	1,295.72	25.88	136.00	231.00	591.25
0X9B90	EOA	11,245,912.26	259,248.29	1,513	60.61	39.39	0.00	7,452.56	171.80	9.00	32.00	95.00
0X55CD	EOA	10,826,096.49	12,631.43	111	19.82	80.18	0.00	97,532.40	113.80	88.00	157.00	281.50
0X882D	SC	8,829,382.76	296,319.33	5,472	52.98	0.22	46.80	1,613.56	54.16	1,141.00	1,344.00	2,318.00
0X48CC	EOA	7,222,030.70	10,662.60	411	100.00	0.00	0.00	17,832.17	26.52	17.00	21.00	25.00
0X612E	EOA	7,199,928.99	100,711.09	606	100.00	0.00	0.00	11,881.07	166.19	80.00	127.50	144.00
0X826A	SC	5,994,600.57	200,206.35	4,420	30.75	0.09	69.16	1,356.24	45.31	1,145.00	1,278.00	1,831.00
0X6226	EOA	5,464,706.66	163,178.78	43,472	99.98	0.02	0.00	125.72	3.76	2.00	4.00	6.00
0XE83F	EOA	5,348,364.74	87,230.70	2,481	45.22	45.67	9.11	2,170.60	35.85	72.00	232.00	928.00
0X65A8	EOA	4,843,464.03	220,693.86	4,221	45.94	45.11	8.96	1,170.48	53.83	64.00	241.00	982.00

[†] - is actually an EOA that was identified via its own SC.

Table 10: Cross-chain arbitrage metrics for arbitrageurs with ≥ 1 % of the total trade volume. EOA - Externally Owned Account, SC - Smart Contract.